

the WiiMote and the Sensor Bar.

The Sensor Bar is about eight inches long and features 10 infrared LEDs, five at each end of the bar. The LEDs farthest away from the centre are pointed slightly away from the centre, the LEDs closest to the centre are pointed slightly inwards, while the rest are pointed straight forward.

The Sensor Bar's cable is almost 12 feet in length – enough to be placed above or below the television, preferably at the centre. If placed above, the sensor should be in line with the front of the television, and if placed below, it should be in line with the front of the surface the television is placed on.

Nintendo notes that it is not necessary to point directly at the Sensor Bar, but pointing significantly away from the bar will disrupt position-sensing ability due to the limited viewing angle of the Wii remote.

The WiiMote can be used accurately in conjunction with the Sensor Bar up to a distance of 16 feet away. The WiiMote's image sensor is used to locate the bar's points of light in the remote's field of view.

The light emitted from each end of the Sensor Bar is focused onto the image sensor which sees the light as two bright dots separated by a distance on the image sensor. The second distance between the two clusters of light emitters in the Sensor Bar is a fixed distance. From these two distances, the Broadway CPU calculates the distance between the WiiMote and the Sensor Bar using triangulation.

In addition, rotation of the WiiMote with respect to the ground can also be calculated from the relative angle of the two dots of light on the image sensor.

The Sensor Bar is required when the WiiMote is controlling up-down, left-right motion of a cursor or reticle (a pointing cursor such as a cross-hair) on the TV screen to point to menu options or objects such as enemies in first-person shooters.

Because the system can also calculate the distance between the WiiMote and the Sensor Bar, the remote can also control slow forward-backward motion of an object in a 3-dimensional game. Rapid forward-backward motion, such as punching in a boxing game, is controlled by the acceleration sensors. Using these acceleration sensors (acting as tilt sensors), the WiiMote can also control rotation of a cursor or other objects.



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